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FUNCTIONAL EQUIVALENCE BETWEEN LIQUIDITY COSTS AND THE UTILITY OF MONEY

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We demonstrate a functional equivalence between using real balances as an argument of the utility function and entering money into liquidity costs which appear in the budget constraint. The liquidity costs can be approximately derived from conventional models of money demand, such as the transactions and precautionary models. We also propose a general class of liquidity cost functions and show an exact duality between it and a broad class of utility functions which include real balances. There is a tendency for the cross-derivative between real balances and the consumption good to be non-negative, though this result will not hold for utility functions which are sufficiently concave.

1. Introduction

An open question in monetary theory is how to best introduce money into optimizing models. One approach has been to simply include real money balances as an argument in the consumer's utility function, on the argument that this represents a 'reduced form' equation in a world of transactions costs or uncertainty. Early examples of this approach are Friedman (1969), Patinkin (1965), Samuelson (1947) and Sidrauski (1967). Models of growth under perfect foresight with money in the utility function have been examined by Brock (1974), Calvo (1979), Fischer (1979), Obstfeld and Rogoff (1983, 1985) and Obstfeld (1984), among others. These authors find that the stability and uniqueness of perfect foresight equilibria are sensitive to properties of the utility function, such as the cross-derivative between money and the consumption good.

The use of real balances in the utility function has been criticized, however, by Clower (1967) and Kareken and Wallace (1980). Clower argues that this approach does not yield a theory where money plays a special role in transactions, and advocates instead the use of 'cash-in-advance' constraints. Karaken and Wallace object to the 'implicit theorizing' which takes place when money is inserted into the utility function, arguing that underlying consistency cannot be checked. In view of these criticisms, it is of substantial

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interest to investigate whether the use of money in the utility function can be explicitly derived from an optimizing model with transaction costs or uncertainty. Fischer (1974) studied this question for money in the production function, and our analysis is complementary to his. An insightful analysis of money in the utility function in the context of overlapping generations models is provided by McCallum (1983).

In this paper we show there exists a functional equivalence between using real balances as an argument of the utility function and entering money into liquidity costs which appear in the budget constraint. A comparison of these two approaches in monetary growth models is given in Dornbusch and Frenkel (1973) and Gray (1984), and the equivalence was first indicated by a simple example in Brock (1974). We demonstrate in section 2 that the liquidity costs appearing in the budget constraint can be approximately derived from conventional models of money demand, such as Baumol (1952), Tobin (1956) and Whalen (1966). We then propose a general class of liquidity cost functions, and show in section 3 an exact duality between it and a broad class of utility functions which include real balances. We characterize the properties of the utility functions in detail. There is a tendency for the cross-derivative between real balances and the consumption good to be non-negative, though this result will not hold for utility functions which are sufficiently concave. Applications of our results are given in section 4, and the proofs of Propositions are gathered in the appendix.

2. Liquidity costs

We shall consider an individual making an intertemporal consumption and savings decision. In period t utility is $U(c_t)$, where c_t denotes consumption and we assume that $U'>0$ and $U''\leq 0$. Denoting real bond and money holdings in period t by b_t and m_t, respectively, the consumer's problem is to choose c_t, $m_t \geq 0$ and b_t to

$$\max \sum_{t=0}^{\infty} \delta^t U(c_t), \tag{1}$$

subject to

$$c_t + \phi(c_t, m_t) + b_t + m_t = w_t + [(1+i_{t-1})b_{t-1} + m_{t-1}](p_{t-1}/p_t), \tag{2}$$

where $0<\delta<1$ is the discount rate.

The budget constraint (2) is understood as follows. On the right side w_t is wage and other non-interest income received at the beginning of period

t, while $[(1+i_{t-1})b_{t-1}+m_{t-1}](p_{t-1}/p_t)$ are real assets carried over from the previous period. Note that the real interest rate r_{t-1} satisfies $(1+r_{t-1}) = (1+i_{t-1})p_{t-1}/p_t$. On the left side $\phi(c_t, m_t)$ are real 'liquidity costs' which the consumer must pay to consume c_t. It is our purpose in this section to justify the role of liquidity costs in the budget constraint, and we now consider three models of money demand which generate these costs.

2.1. Transactions model

First, we consider the transactions model of Baumol (1952) and Tobin (1956). We shall suppose that *within* each discrete time period t the individual consumes at the *flow* rate $\bar{c}_t$, which is constant within periods but can vary across them. Prices are constant during each period. The consumption $\bar{c}_t$ is financed by money holdings, which are obtained by converting bonds to money at the transactions cost of k. Suppose that the individual makes N_t of these transactions within the period t. To finance the continuous consumption of $\bar{c}_t$ with a minimum of forgone interest, the transactions will be evenly spaced and the individual will obtain $\mu_t = \bar{c}_t/N_t$ money balances at each. Then average money holdings over the period are simply $\overline{m}_t = \mu_t/2 = \bar{c}_t/2N_t$ and total transactions costs are $kN_t = k\bar{c}_t/2\overline{m}_t$.

Next, we should check the individual's budget constraint during period t. In the discrete-time constraint (2) we have specified that the principal and real interest from bonds in period $t-1$ is carried over to period t. However, with the level of bonds changing within each period, we should instead specify that the principle from *end-of-period* bonds and the real interest from *average* bond holdings is carried over to the next period. It is cumbersome to have a budget constraint involving two different levels of bonds. So we shall now show that the budget constraint for the Baumol–Tobin model can be written solely in terms of average bond holdings.

To this end, let y_t denote initial assets in period t. Since the consumer obtains μ_t money from each transaction at the cost of k, average bond holdings are

$$\bar{b}_t = y_t - (1/N_t)\left[(\mu_t + k) + 2(\mu_t + k) + \cdots + N_t(\mu_t + k)\right]. \tag{3}$$

The summed terms on the right side of (3) are the successive reductions in bond holdings. Using results from above, this expression is simplified as

$$\begin{aligned} y_t &= \bar{b}_t + (\mu_t + k)(N_t + 1)/2 \\ &= \bar{b}_t + \overline{m}_t + (\bar{c}_t + kN_t)/2 + k/2. \end{aligned} \tag{3'}$$

Next, we can compute y_t, assets at the beginning of period t. In the previous period the consumer spent $\bar{c}_{t-1} + kN_{t-1}$ and received real interest income of

$r_{t-1}\bar{b}_{t-1}$. It follows that

$$\begin{aligned} y_t &= w_t + (y_{t-1} - \bar{c}_{t-1} - kN_{t-1}) + r_{t-1}\bar{b}_{t-1} \\ &= w_t + (1 + r_{t-1})\bar{b}_{t-1} + \bar{m}_{t-1} - (\bar{c}_{t-1} + kN_{t-1})/2 + k/2, \end{aligned} \tag{4}$$

using (3′) evaluated for $t-1$. Then we can equate (4) to (3′), obtaining

$$\begin{aligned} &[\bar{c}_t + \phi(\bar{c}_t, \bar{m}_t)]/2 + \bar{b}_t + \bar{m}_t \\ &= w_t + (1 + r_{t-1})\bar{b}_{t-1} + \bar{m}_{t-1} - [\bar{c}_{t-1} + \phi(\bar{c}_{t-1}, \bar{m}_{t-1})]/2, \end{aligned} \tag{5}$$

where $\phi(\bar{c}_t, \bar{m}_t) = kN_t = k\bar{c}_t/2\bar{m}_t$ are the liquidity costs.

Our final step in deriving the Baumol–Tobin budget constraint is to allow prices to vary across periods. For simplicity, we shall assume that inflation is literally paid as a tax at the end of each period. That is, with prices changing from p_{t-1} to p_t the amount $\bar{m}_{t-1}[1 - (p_{t-1}/p_t)]$ is withheld from the consumer at the end of period $t-1$ and subtracted from the right of (5). The budget constraint is then

$$\begin{aligned} &[\bar{c}_t + \phi(\bar{c}_t, \bar{m}_t)]/2 + \bar{b}_t + \bar{m}_t \\ &= w_t + [(1 + i_{t-1})\bar{b}_{t-1} + \bar{m}_{t-1}](p_{t-1}/p_t) \\ &\quad - [\bar{c}_{t-1} + \phi(\bar{c}_{t-1}, \bar{m}_{t-1})]/2. \end{aligned} \tag{5′}$$

Under this assumption, the consumer need not obtain higher nominal balances to finance consumption at rising prices, but rather just realizes that an inflation tax will be levied in proportion to average real balances $\bar{m}_{t-1}$. Without the assumption, liquidity costs would depend on the inflation rate.[1] The consequences of this assumption are considered in section 4 when we compare our model to Jovanovic (1982). In general, writing liquidity costs as a function of only consumption and real balances may require some simplifying assumption, as we shall also find below for the precautionary model.

Summing up, we have shown that the Baumol–Tobin model gives rise to well defined liquidity costs, with the budget constraint (5′). Note that this budget constraint is very similar in form to (2).[2] The liquidity costs $\phi(\bar{c}_t, \bar{m}_t)$

[1] For example, suppose that between each bond-to-money transaction prices rise linearly at the rate π_t. Then the real balances needed to finance consumption in the interval are $\mu_t = \int_0^{1/N_t} \bar{c}_t(1 + \pi_t s)\,ds = (\bar{c}_t/N_t)(1 + \pi_t/2N_t)$. Solving for N_t, we obtain $kN_t = k(\bar{c}_t/2\mu_t)[1 + (1 + 2\mu_t\pi_t/\bar{c}_t)^{1/2}]$, so the liquidity costs depend on the inflation rate.

[2] Aside from notation, the only difference between these budget constraints is that in (5′) the forgone real interest from increasing consumption by one unit in the current period is $r_t/2$, whereas in (2) it is r_t. For expositional ease we shall use (2) as a generic form of the budget constraint, while showing how our analysis applies to (5′) in footnote 7.

$= k\bar{c}_t/2\bar{m}_t$ are increasing in consumption, decreasing in money, and *homogeneous of degree zero*: multiplying $\bar{c}_t$ and $\bar{m}_t$ by the factor $\lambda > 0$ will not affect ϕ. This reflects our assumption that the cost of converting bonds to money is independent of the transaction size. By treating ϕ as a continuous function of $\bar{c}_t$ and $\bar{m}_t$ we are implicitly permitting the number of transactions N_t to take on non-integer values. We next consider a generalization of the transactions model due to Barro (1976), in which the integer constraints are recognized.

2.2. Generalized transactions model

Barro (1976) considers persons making transactions between bonds and money, as described above, but permits N_t to only take on integer values. Then the response of money demand to a change in income or the interest rate depends critically on whether or not an individual makes a discrete change in N_t. Barro overcomes this 'discontinuous' behaviour by aggregating over the population, using a gamma distribution for income. His results are summarized as follows.

We let $\bar{c}_t$ denote the expected flow consumption rate over the population, $\bar{N}_t$ denote the expected number of transactions, and $\bar{m}_t$ denote the expected money holdings. Then Barro finds that[3]

$$\bar{m}_t = \bar{c}_t g(\bar{N}_t)/2, \tag{6}$$

where g is a decreasing function of $\bar{N}_t$ with an elasticity lying between zero and unity and decreasing in $\bar{N}_t$. This formula may be contrasted with the Baumol–Tobin model where $\bar{m}_t = \bar{c}_t/2N_t$. With a cost of k per transaction between bonds and money, the expected liquidity costs over the population are $\phi(\bar{c}_t, \bar{m}_t) = k\bar{N}_t$. Then from (6) we can write liquidity costs as

$$\phi(\bar{c}_t, \bar{m}_t) = kg^{-1}(2\bar{m}_t/\bar{c}_t) \equiv h(\bar{c}_t/\bar{m}_t). \tag{7}$$

In the Baumol–Tobin model we found that liquidity costs were a linear function of $\bar{c}_t/\bar{m}_t$, and we now find that in the Barro formulation liquidity costs are a non-linear function. Moreover, multiplying both $\bar{c}_t$ and $\bar{m}_t$ by the factor $\lambda > 0$ will not affect ϕ: liquidity costs in this generalized transactions model are still *homogeneous of degree zero*. We shall impose the following

[3]Eq. (6) is obtained from eq. (14) of Barro (1976) with $m_t = \mathrm{E}\hat{M}$, $c_t = \bar{y}_t$, $T = 1$ and $g(\bar{N}_t) = \Phi(\bar{n})$. Note that Barro defines $\bar{N}_t = \bar{n}$ as the non-integer number of transactions which an individual with the *mean* consumption rate desires. It is related to the *expected number* of transactions $\mathrm{E}N_t$, computed by recognizing the integer constraints and integrating over the population, according to $\mathrm{E}N_t = \sum_{i=1}^{\infty}[1 + 2i(i-1)/\bar{N}_t^2]\exp[-2i(i-1)/\bar{N}_t^2]$. If $\bar{N}_t = 1, 2, 3, 5, 10$, respectively, then we can compute that $\mathrm{E}N_t = 1.09, 1.95, 2.87, 4.73, 9.45$. We shall treat $\bar{N}_t$ and $\mathrm{E}N_t$ interchangeably.

assumptions on h:

$$h(0)=0, \quad h'>0, \quad h''\geq 0,$$

$$(\mathrm{d}/\mathrm{d}z)[h'(z)z/h(z)]\leq 0. \tag{8}$$

These assumptions state that liquidity costs are an increasing, convex function of $\bar{c}_t/\overline{m}_t$, with a constant or diminishing elasticity, and are consistent with the results of Barro on the function g. The assumptions (8) are satisfied, for example, if $h(\bar{c}_t/\overline{m}_t)=k(\bar{c}_t/\overline{m}_t)^a$ with $a\geq 1$.

2.3. *Precautionary model*

In the precautionary demand model of Whalen (1966) money is held to finance consumption and there is a penalty cost associated with a cash shortfall. Whalen considers two forms of penalty cost: constant costs and costs proportional to the cash shortfall. The case of constant penalty costs, when used in our analysis below, would lead to liquidity costs similar to the generalized transactions model already considered. So instead, we shall now examine the case of *proportional* penalty costs.

We shall suppose that the individual faces uncertainty in the price of the consumption good but must make several decisions before the uncertainty is resolved. First, assets must be allocated between bonds and money ex-ante. In addition, a consumption decision of c_t must be made ex-ante. Then $(1+\theta)c_t$ denotes ex-post real consumption expenditures, where θ is the random variable denoting price uncertainty with $\mathrm{E}\theta=0$ and $\theta>-1$. Let m_t denote planned real balances at the end of period t, that is, the real balances remaining if $\theta=0$ and consumption expenditures are c_t.

The choice of m_t will affect the probability of a cash shortfall. In particular, if $\theta c_t\leq m_t$, then the precautionary balances m_t are sufficient to cover the unexpected expenditure θc_t. However, if $\theta c_t>m_t$, then a cash shortfall occurs. We shall suppose that the consumer covers the cash shortfall by purchasing that portion of the good on 'credit'. Credit is extended at the interest cost of ρ, so total liquidity costs are $\rho(\theta c_t-m_t)$. We shall assume that liquidity costs are paid in the current period. Whether a cash shortfall occurs or not, the ex-post real balances at the end of the period are $\tilde{m}_t=m_t-\theta c_t$, where $\tilde{m}_t<0$ indicates that credit has been extended.

Turning to the individuals budget constraint, let $\tilde{b}_t$ denote ex-post real bond holdings. Define liquidity costs as

$$\begin{aligned}\tilde{\phi}(c_t,m_t)&=0 &&\text{if}\quad \theta\leq m_t/c_t,\\ &=\rho(\theta c_t-m_t) &&\text{if}\quad \theta>m_t/c_t.\end{aligned} \tag{9}$$

Then we can write the budget constraint as

$$\begin{aligned}\tilde{b}_t &= \tilde{y}_t - (1+\theta)c_t - \tilde{\phi}(c_t, m_t) - \tilde{m}_t \\ &= \tilde{y}_t - c_t - \tilde{\phi}(c_t, m_t) - m_t,\end{aligned} \tag{10}$$

since $\tilde{m}_t = m_t - \theta c_t$. In this budget constraint $\tilde{b}_t$ is random since liquidity costs are also random, while c_t and m_t are non-stochastic choice variables of the consumer.

The budget constraint (10) is equivalent to (2) except for its stochastic nature. To proceed further, we shall assume that the consumer maximizes utility subject to the *expected value* of (10). This approach should be considered an approximation and allows us to ignore certain terms involving the variance of assets and risk aversion.[4] These terms would otherwise be part of the consumer's liquidity costs. Under the expected value approach the budget constraint (10) is equivalent to (2) with the definitions $b_t \equiv \mathrm{E}\tilde{b}_t$, $\phi \equiv \mathrm{E}\tilde{\phi}$ and $y_t \equiv \tilde{y}_t$. Liquidity costs are now computed as

$$\phi(c_t, m_t) \equiv \mathrm{E}\tilde{\phi} = \int_{m_t/c_t}^{\bar{\theta}} \rho(\theta c_t - m_t) f(\theta)\,\mathrm{d}\theta, \tag{11}$$

where $f(\theta)$ is the density function and $\bar{\theta}$ is the upper bound of θ.

It is readily seen from (11) that, for $m_t/c_t < \bar{\theta}$, transactions costs are increasing in consumption, decreasing in real balances, and *homogeneous of degree one*: for $\lambda > 0$, $\phi(\lambda c_t, \lambda m_t) = \lambda\phi(c_t, m_t)$. This result depends on our assumption of a *proportional* penalty of cash shortfall. In the alternative case of a constant penalty the liquidity costs would be homogeneous of degree zero, as for the generalized transactions model analysed above. Note that from (11) a change in the variance or higher-order moments of θ will shift the function ϕ. Thus, in solving the consumer's problem for given liquidity costs, we are implicitly assuming that the distribution of prices is constant. This restriction on the precautionary model will also apply when money is used in the utility function.

2.4. *General properties of liquidity costs*

We have shown that the generalized transactions and precautionary models can be stated in terms of liquidity costs appearing in the consumer's budget

[4] Let $\mathrm{E}J_t(\tilde{y})$ denote the optimized value of expected discounted utility from period t onwards, subject to the stochastic budget constraint (10). Then we have $\mathrm{E}J_t(\tilde{y}_t) = J_t(\mathrm{E}\tilde{y}_t) + \frac{1}{2}J_t''(\mathrm{E}\tilde{y}_t)\mathrm{var}(\tilde{y}_t)$. In maximizing utility subject to the expected budget constraint we are ignoring the term $J_t''\,\mathrm{var}(\tilde{y}_t)$, which will generally be non-zero. Since the ignored term is time-dependent omitting it is a great simplification.

constraint. Treating these two models as important special cases, it is clear that other liquidity cost functions could be obtained from models which we have not examined [e.g., the 'shopping time' specification of McCallum (1983)]. We now propose the following general set of assumptions on liquidity costs:

Assumption 1. For all $c_t \geq 0$, $m_t > 0$, the liquidity costs $\phi(c_t, m_t)$ are twice continuously differentiable and satisfy:

(a) $\phi \geq 0$, $\phi(0, m_t) = 0$;

(b) $\phi_c \geq 0$, $\phi_m \leq 0$;

(c) $\phi_{cc} \geq 0$, $\phi_{mm} \geq 0$, $\phi_{mc} \leq 0$;

(d) $c_t + \phi(c_t, m_t)$ is quasi-convex with expansion paths of non-negative slope.

The condition $\phi(0, m_t) = 0$ states that liquidity costs only occur due to positive consumption. Assumption 1(b) is straightforward, and Assumption 1(c) adds further structure with constant or rising marginal costs of consumption and constant or diminishing marginal benefits from real balances. Higher real balances lead to unchanged or lower marginal consumption costs.

Assumption 1(d) is understood as follows. In figs. 1 and 2, respectively, we show the iso-curves of $\phi(c_t, m_t)$ for the generalized transactions and precautionary models. In the former case the iso-curves are simply rays through the

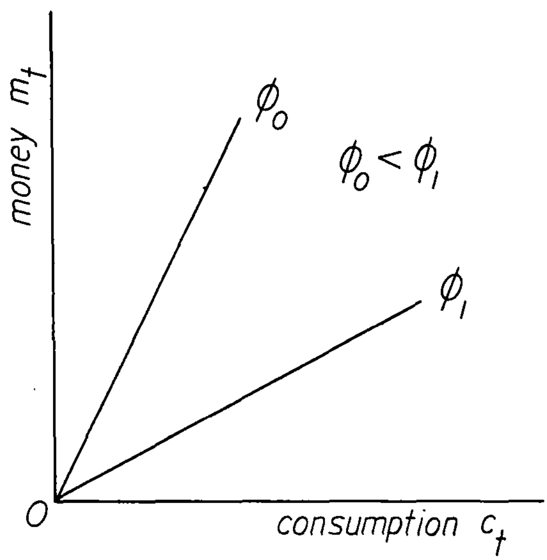

Fig. 1. Iso-curves of liquidity costs for the generalized transactions model.

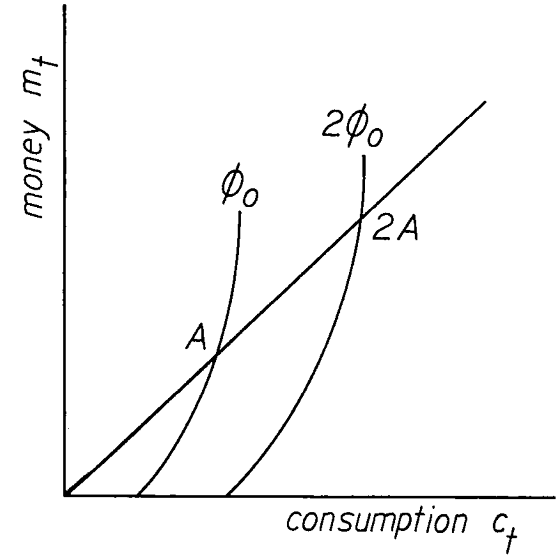

Fig. 2. Iso-curves of liquidity costs for the precautionary model.

origin since $\phi(c_t, m_t) = h(c_t/m_t)$. For the precautionary model it can be verified that the iso-curves have rising slope as shown and the *expansion paths*, joining points of equal slope on the iso-curves, are rays through the origin.[5] In both cases, the constant or rising slope of the iso-curves means that the function ϕ is *quasi-convex*. It can also be verified that the function $c_t + \phi(c_t, m_t)$ is quasi-convex with upward sloping expansion paths. Then Assumption 1(d) applies these requirements to the general class of liquidity cost functions. We leave it to the reader to verify that *any sum of liquidity costs from the generalized transactions model [(7) and (8)] and the precautionary model (11) satisfies Assumption 1.*

3. Utility of money

Having examined liquidity costs in some detail, we can now demonstrate the equivalence between liquidity costs in the budget constraint and entering money into the utility function. We first restate the consumer's problem as that of choosing $x_t, m_t \geq 0$ and b_t to

$$\max \sum_{t=0}^{\infty} \delta^t V(x_t, m_t), \tag{12}$$

[5] From (11) we obtain $-\phi_c/\phi_m = [\int_{m_t/c_t}^{\bar{\theta}} \theta f(\theta)\,\mathrm{d}\theta]/[\int_{m_t/c_t}^{\bar{\theta}} f(\theta)\,\mathrm{d}\theta]$. This expression depends only on the ratio m_t/c_t and is an increasing function for $m_t/c_t < \bar{\theta}$, $f(m_t/c_t) \neq 0$.

subject to

$$x_t + b_t + m_t = y_t, \tag{13}$$

where y_t is the right side of (2). In this statement of the problem $V(x_t, m_t)$ denotes utility, depending on real consumption x_t and real money balances m_t.

It is evident that (12) is equivalent to (1) if, for all c_t, m_t and x_t,[6]

$$U(c_t) \equiv V(x_t, m_t). \tag{14}$$

Next, we see that the budget constraints (2) and (13) are equivalent if[7]

$$c_t + \phi(c_t, m_t) \equiv x_t. \tag{15}$$

We interpret (15) as a definition of 'gross consumption' x_t, including both 'net consumption' c_t and liquidity costs. Substituting (15) into (14) we obtain

$$U(c_t) \equiv V\left[c_t + \phi(c_t, m_t), m_t\right]. \tag{16}$$

We shall say the functions (U, ϕ) are *equivalent* to V, and conversely, if (16) holds for all c_t and m_t. This means that problems (1) and (12) differ only in their functional notation. Note that we have defined equivalence directly in terms of the liquidity cost and utility functions, without reference to the optimal programs for problems (1) and (12). This definition is in the spirit of the duality between production and cost functions, where the duality is defined directly in terms of these functions and without reference to the optimal input demands. When our definition of equivalence holds, then (c_t^*, m_t^*, b_t^*) solves (1) if and only if (x_t^*, m_t^*, b_t^*) solves (12), where $c_t^* + \phi(c_t^*, m_t^*) \equiv x_t^*$.

It will be convenient to have another definition of equivalence, which does not depend on U. Defining $W(x_t, m_t) = U^{-1}[V(x_t, m_t)]$, we can rewrite (16) as

$$c_t \equiv W\left[c_t + \phi(c_t, m_t), m_t\right]. \tag{16'}$$

We shall say that ϕ is *equivalent* to W, and conversely, if (16′) holds for all c_t and m_t. Note that, given the utility function W, then V is simply obtained as a the concave transformation $V(x_t, m_t) = U[W(x_t, m_t)]$ with $U' > 0$, $U'' \leq 0$.

[6] To satisfy the definition of equivalence (14)–(16′) should hold for all $c_t, x_t \geq 0$ and $m_t > 0$. We omit $m_t = 0$ from consideration since in that case liquidity costs $\phi(c_t, m_t)$ may not be well-defined, as with $\phi(\bar{c}_t, \overline{m}_t) = k\bar{c}_t/2\overline{m}_t$ in the Baumol–Tobin model, for example.

[7] For the transactions model the budget constraint (13) should be written as $\bar{x}_t/2 + \bar{b}_t + \overline{m}_t = w_t + [(1 + i_{t-1})\bar{b}_{t-1} + \overline{m}_{t-1}](p_{t-1}/p_t) - \bar{x}_{t-1}/2$. This budget constraint is equivalent to (5′) if (15) holds.

We propose the following general conditions on W:

Assumption 2. For all $x_t \geq 0$, $m_t > 0$, the utility function $W(x_t, m_t)$ is twice continuously differentiable and satisfies:

(a) $W \geq 0$, $W(0, m_t) = 0$, $W(x_t, m_t) \to \infty$ as $x_t \to \infty$ for fixed $m_t > 0$;

(b) $W_m \geq 0$, $0 < W_x \leq 1$;

(c) $W_{xx} \leq 0$, $W_{mm} \leq 0$, $W_{xm} \geq 0$;

(d) W is quasi-concave with Engel curves of non-negative slope.

The condition $W(0, m_t) = 0$, which implies $W_m(0, m_t) = 0$, says that money is not valued in utility unless consumption occurs. The condition $W_x \leq 1$ means only that the marginal utility of consumption is bounded from above; the upper bound can then be set at unity by choice of units for W. It is significant that we assume $W_{xm} \geq 0$ since the sign of this derivative affects the properties of monetary equilibria. Since V is obtained as an increasing, concave transformation of W, it will satisfy $V_m \geq 0$, $V_x > 0$, $V_{xx} \leq 0$, $V_{mm} \leq 0$ and Assumption 2(d). Applying this concave transformation can, however, result in $V_{xm} < 0$ as will be demonstrated below.

We now state our principal equivalence result, proved in the appendix:

Proposition 1. If the liquidity costs $\phi(c_t, m_t)$ satisfy Assumption 1, then there exists an equivalent utility function $W(x_t, m_t)$ satisfying Assumption 2, and conversely.

An essential part of this proof is that given the liquidity costs ϕ, the utility function W is simply obtained by inverting (15) to obtain 'net' consumption depending on 'gross' consumption and real balances: $c_t = W(x_t, m_t)$. Thus, W can also be interpreted as a household production function which specifies the level of 'net' consumption. While W is quasi-concave, it need not be concave, as shown below for the generalized transactions model. In some cases the non-concavity of W may be overcome when the transformation U is applied, so that $V(x_t, m_t) = U[W(x_t, m_t)]$ is concave. However, in general the concavity of V must be imposed as an additional assumption.

To investigate the sign of V_{xm}, we can differentiate (16) and rearrange terms to obtain

$$V_{xm} = \left[\phi_m U'/(1+\phi_c)^2 c_t\right]\left[R_u + \phi_{cc}c_t/(1+\phi_c) - \phi_{cm}c_t/\phi_m\right], \qquad (17)$$

where $R_u = -c_t U''/U' \geq 0$ is the index of relative risk aversion of U. The presence of this term tends to make (17) negative, recalling that $\phi_m \leq 0$. Thus, for utility functions which are sufficiently concave, the cross-derivative V_{xm} will be negative over some region. The difference between the last two terms on the right of (17) is non-positive from Assumption 1(d), so the entire expression is non-negative if $R_u = 0$.

Having identified the general class of utility functions which are equivalent to well behaved liquidity costs, we can now consider special cases. We first consider the generalized transactions model.

Proposition 2. Suppose that ϕ is obtained from the generalized transactions model. Then:

(*a*) *The Engel curves of the equivalent utility function have a smaller slope relative to the x_t axis than a ray through the origin*;

(*b*) *W is not concave, but V is concave if $R_u \geq \frac{1}{2}$.*

The Engel curves of V and W are illustrated in fig. 3, points A and B, where they cross from above rays through the origin. The result that W is not concave reflects the 'increasing returns' which are inherent in the transactions model, whereby the cost of converting bonds to money is independent of the transactions size. In this case if U is sufficiently concave, $R_u \geq \frac{1}{2}$, then applying this concave transformation will ensure that V is also concave.

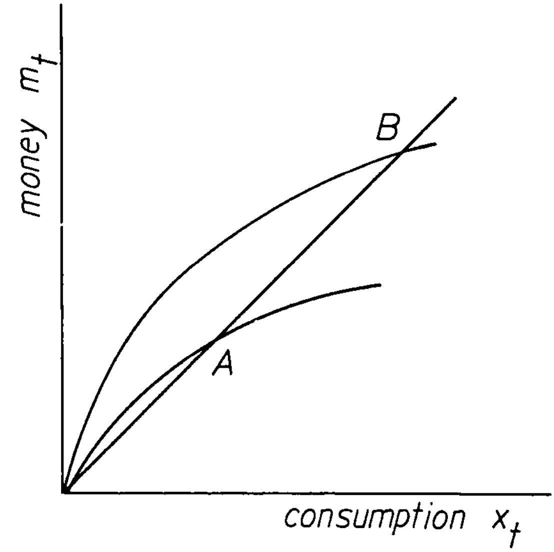

Fig. 3. Engel curves of the utility function for the generalized transactions model.

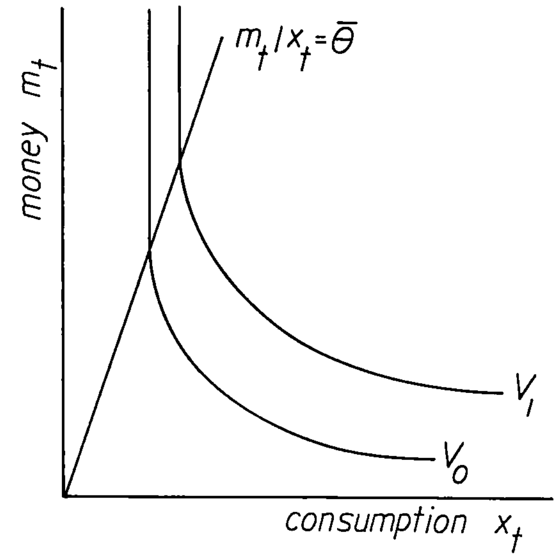

Fig. 4. Indifference curves of the utility function for the precautionary model.

We next consider the Baumol–Tobin model as a special case of Proposition 2 and obtain an explicit solution for the utility function. For the Baumol–Tobin model liquidity costs are $kc_t/2m_t$. Then $x_t = c_t + kc_t/2m_t$, so $c_t = x_t/(1 + k/2m_t)$. The utility function is then $V(x_t, m_t) = U(c_t) = U[x_t/(1 + k/2m_t)]$. If $U(c_t) = \ln c_t$ we obtain $V = \ln x_t - \ln(1 + k/m_t)$, so that V is *separable* between goods and money. More generally, $V_{xm} > (<)\ 0$ as $R_u < (>)\ 1$.

Finally, we consider special results for the precautionary model of section 2.3. It is readily shown from (16′) that the liquidity cost function is homogeneous of degree one if and only if W also satisfies this property. We now demonstrate that many utility functions of this type can be derived from the precautionary model.

Proposition 3. Let W satisfy Assumption 2 and be homogeneous of degree one, with (i) $W_m = 0$ for $m_t/x_t \geq \bar{\theta}$, (ii) $W_m > 0$ and $W_{mm} < 0$ for $m_t/x_t < \bar{\theta}$, (iii) W_m/W_x is bounded above. Then there exists a liquidity cost function obtained from the precautionary model with proportional penalty which is equivalent to W.

The condition $W_m = 0$ for $m_t/x_t \geq \bar{\theta}$ is illustrated in fig. 4, where the indifference curves are vertical in this range. For the precautionary model a zero marginal utility of money occurs when m_t/x_t exceeds the highest unexpected rise in prices $\bar{\theta}$, so that credit is not used and liquidity costs are zero. This situation of a zero marginal utility has been used by Friedman (1969) to obtain the optimum quantity of money.

To understand the generality of Proposition 3, note that $\bar{\theta}$ can be chosen as infinity in condition (i), while (ii) is quite conventional, and (iii) affects the function W only in a neighborhood of $m_t = 0$. Thus, nearly the entire homogeneous of degree one class for W can be obtained from the precautionary model. This class of utility functions has been used by Dornbusch and Mussa (1975). Note that since W is quasi-concave and homogeneous of degree one, it is concave. Then the utility function V obtained as $V(x_t, m_t) = U[W(x_t, m_t)]$ with $U' > 0$, $U'' \leq 0$, is homothetic and concave. Since nearly any homogeneous of degree one W can be obtained from the precautionary model, it follows that nearly all homothetic and concave utility functions V can be so obtained.

4. Applications

Our analysis can be used to unify a number of results in the literature:

(a) Calvo (1979) and Obstfeld (1984) investigate multiple stable equilibria in a monetary growth model, where the former uses real balances in a production function $h(m_t)$ and the latter uses real balances in the utility function. Treating the production function of Calvo as negative liquidity costs, their budget constraints are identical if $c_t - h(m_t) = x_t$. Then (16) becomes $U(c_t) = V[c_t - h(m_t), m_t]$. Differentiating this condition and rearranging, we obtain $m_t V_{xm}/V_x = m_t h' U''/U'$. The condition derived by Obstfeld to rule out multiple stable equilibria is that the left side of this expression exceed -1, whereas the condition of Calvo is that the right side exceed -1, so these are equivalent.

(b) Jovanovic (1982) examines a general equilibrium Baumol–Tobin model, with transaction costs modelled as in section 2.1. It follows that his results should be equivalent to using money in the utility function, as in Sidrauski (1967). When comparing these papers there are two apparent inconsistencies. First, in Sidrauski steady-state consumption is independent of the inflation rate, whereas this result does not hold in Jovanovic. This difference arises since in the former consumption is the 'gross' form x_t, while in the latter it is the 'net' form c_t. Second, in Sidrauski the steady-state capital stock is independent of inflation, whereas Jovanovic finds that the capital stock changes with inflation (the direction of change is ambiguous). The only essential difference between our version of the Baumol–Tobin model and that of Jovanovic is that we have treated inflation as a tax paid at the end of each period (see section 2.1). This approximation should become negligible as the *holding period* between each bond-to-money transaction becomes arbitrarily small. Indeed, it can be shown in Jovanovic's model that as the holding period approaches zero, the change in the economy's capital stock with respect to steady-state inflation also approaches zero. In this local sense, the capital stock is independent of inflation.[8]

(c) Scheinkman (1980) and Obstfeld and Rogoff (1983) have shown that hyperinflation equilibria, with real balances approaching zero, can be ruled out if and only if $\lim_{m_t \to 0} m_t V_m / V_x > 0$. From (16) we have that $-\phi_m = V_m / V_x$, so the analogous condition stated in terms of liquidity costs is $\lim_{m_t \to 0} m_t \phi_m < 0$, as obtained by Gray (1984). Using liquidity costs does not help to restrict the possibility of hyperinflation equilibria, since the latter condition may or may not hold under Assumption 1.

More recently, Obstfeld and Rogoff (1985) have shown that a necessary condition to obtain implosive equilibria, with the price level approaching zero, is that $\lim_{m_t \to \infty} V(x_t, m_t) = \infty$. This condition can never hold if the utility function is derived from liquidity costs satisfying Assumption 1, since it violates $W(0, m_t) = 0$ in Assumption 2.

(d) Clower (1967) proposed the use of the 'cash-in-advance' constraint

$$c_t \leq m_{t-1} p_{t-1} / p_t, \tag{18}$$

which states that nominal consumption in the current period cannot exceed nominal money balances carried over from the previous period. Adding this constraint to the consumer's problem (1) is equivalent to using the following Leontief utility function in (12):

$$V(x_t, m_{t-1} p_{t-1} / p_t) = U(\min\{x_t, m_{t-1} p_{t-1} / p_t\}). \tag{19}$$

Thus, cash-in-advance constraints are an important special case of a utility function which includes real balances. The crucial feature of this utility function is its zero elasticity of substitution between goods and money.[9] [Note that the price ratio as an argument of V would not occur in a continuous-time version of cash-in-advance constraints, as in Feenstra (1985).] We shall now argue that the zero elasticity of substitution means that (19) is approximated by utility functions with a *positive* cross-derivative between goods and money, regardless of how concave U may be.

[8] The derivation of this result is available on request. Barro and Fischer (1976) note several reasons why the superneutrality results of Sidrauski are not robust.

[9] This result may be contrasted with Lucas and Stokey (1983, fn. 7). They deal with an economy with 'cash' goods and 'credit' goods, where only the former are subject to (18). Define total consumption as the sum of cash and credit goods purchased. Then the derived utility function can have a non-zero elasticity of substitution between total consumption and real balances held for cash goods. However, the elasticity of substitution between cash goods and real balances is zero.

This result is obtained as follows. For any utility function $V(x_t, m_t) = U[W(x_t, m_t)]$, where W is homogeneous of degree one, we have by differentiation, $V_{xm} = (U'W_xW_m/W)[(1/\sigma_{xm}) - R_u]$, where $\sigma_{xm} = W_xW_m/W_{xm}W \geq 0$ is the elasticity of substitution between x_t and m_t and $R_u = -c_tU''/U' \geq 0$. For any fixed value of R_u, as $\sigma_{xm} \to 0$, then V_{xm} becomes positive. Thus, we can consider approximating (19) by utility functions with the same values of R_u, but successively smaller elasticities of substitution, and this sequence of functions must eventually have positive cross-derivatives between goods and money.

We feel this result is very significant, since it is generally known that the sign of V_{xm} affects the properties of monetary equilibria. From this result, *we can conjecture that models with cash-in-advance constraints like (18) will display similar qualitative properties to those with* $V_{xm} > 0$. This conjecture applies to the non-neutrality results derived by Rotemberg (1982), for example, using the cash-in-advance constraint. It would not apply to the model of Stockman (1981), however, since he assumes that money is needed for purchases of consumption and bonds which differs from (18).

Appendix[10]

A.1. Proof of Proposition 1

To prove (a), suppose that ϕ satisfying Assumption 1 is given. Then define $x_t \equiv c_t + \phi(c_t, m_t)$ as in (15). Note that from Assumption 1(a) the range of x_t is $[0, \infty)$ for c_t in that interval and $m_t > 0$ fixed. Since $1 + \phi_c > 0$, we can invert (15) to obtain $c_t = W(x_t, m_t)$, as in (16′). To check that Assumptions 2(a) and 2(b) are satisfied, we compute from the inverse function theorem,

$$W(0, m_t) = 0, \tag{A.1}$$

$$W_x = 1/(1 + \phi_c), \tag{A.2}$$

$$W_m = -\phi_m/(1 + \phi_c), \tag{A.3}$$

so $0 < W_x \leq 1$ and $W_m \geq 0$. Since ϕ is a non-decreasing, continuous function of c_t, if $x_t = c_t + \phi(c_t, m_t) \to \infty$ for fixed $m_t > 0$, then this implies $c_t = W \to \infty$.

Conversely, beginning with W satisfying Assumption 2, define $c_t \equiv W(x_t, m_t)$ as in (16′). Note that from Assumption 2(a) the range of c_t is $[0, \infty)$ for x_t in that interval and $m_t > 0$ fixed. Since $W_x > 0$, we can invert $c_t =$

[10] Longer versions of these proofs are contained in the author's working paper, available on request.

$W(x_t, m_t)$ to obtain $x_t = \chi(c_t, m_t)$. Then we define $\phi(c_t, m_t) \equiv x_t - c_t = \chi(c_t, m_t) - c_t$, as in (15). To check Assumptions 1(a) and 1(b) we compute

$$\phi(0, m_t) = \chi(0, m_t) = 0, \tag{A.4}$$

$$\phi_c = \chi_c - 1 = (1/W_x) - 1 \geq 0, \tag{A.5}$$

$$\phi_m = \chi_m = -(W_m/W_x) \leq 0, \tag{A.6}$$

and it follows by integrating along a path of constant m_t that $\phi(c_t, m_t) \geq 0$.

We next consider the relations between second derivatives of W and ϕ.

(i) Differentiating (A.2) with respect to c_t,

$$0 = W_{xx}(1 + \phi_c)^2 + W_x\phi_{cc},$$

so

$$W_{xx} = -W_x\phi_{cc}/(1 + \phi_c)^2.$$

(ii) Differentiating (A.2) with respect to m_t, holding $x_t = c_t + \phi$ fixed,

$$0 = W_{xm}(1 + \phi_c) + W_x \frac{\mathrm{d}}{\mathrm{d}m}\phi_c\bigg|_{c+\phi},$$

where

$$\frac{\mathrm{d}}{\mathrm{d}m}\phi_c\bigg|_{\bar{c}+\phi} = \phi_{cm} - \frac{\phi_{cc}\phi_m}{1 + \phi_c} = (1 + \phi_c)\frac{\partial}{\partial c}\left(\frac{\phi_m}{1 + \phi_c}\right).$$

Thus,

$$W_{xm} = -W_x\frac{\partial}{\partial c}\left(\frac{\phi_m}{1 + \phi_c}\right).$$

(iii) Differentiating (A.6) with respect to x_t,

$$\frac{\partial}{\partial x}\left(\frac{-W_m}{W_x}\right) = \phi_{mc}\frac{\mathrm{d}c}{\mathrm{d}x}\bigg|_m = \frac{\phi_{mc}}{1 + \phi_c}.$$

(iv) Differentiating (A.6) with respect to m_t, holding $x_t = c_t + \phi$ fixed,

$$\frac{\partial}{\partial m}\left(\frac{-W_m}{W_x}\right) = \frac{\mathrm{d}}{\mathrm{d}m}\phi_m\bigg|_{c+\phi} = \phi_{mm} - \frac{\phi_{mc}\phi_m}{1+\phi_c}$$

$$= (1+\phi_c)\frac{\partial}{\partial m}\left(\frac{\phi_m}{1+\phi_c}\right).$$

(v) Using (ii) and (iv),

$$\frac{\partial}{\partial c}\left(\frac{\phi_m}{1+\phi_c}\right) - \left(\frac{1+\phi_c}{\phi_m}\right)\frac{\partial}{\partial m}\left(\frac{\phi_m}{1+\phi_c}\right)$$

$$= -\frac{W_{xm}}{W_x} + \left(\frac{W_x}{W_m}\right)\frac{\partial}{\partial m}\left(\frac{-W_m}{W_x}\right) = -\frac{W_{mm}}{W_m},$$

since

$$\frac{\partial}{\partial m}\left(\frac{-W_m}{W_x}\right) = -\frac{W_x W_{mm} - W_m W_{xm}}{W_x^2}.$$

(vi) Differentiating (A.6) with respect to m_t, holding c_t fixed,

$$\phi_{mm} = \frac{\mathrm{d}}{\mathrm{d}m}\left(\frac{-W_m}{W_x}\right)\bigg|_c = \frac{\partial}{\partial m}\left(\frac{-W_m}{W_x}\right) + \phi_m\frac{\partial}{\partial x}\left(\frac{-W_m}{W_x}\right).$$

Equations (i)–(vi) show that the remaining conditions in Assumption 1 imply the conditions of Assumption 2, and conversely. Q.E.D.

A.2. Proof of Proposition 2

To prove (a), differentiate (16′) with respect to m_t, obtaining

$$\frac{W_m[c_t + h(c_t/m_t), m_t]}{W_x[c_t + h(c_t/m_t), m_t]} = -\phi_m = \frac{c_t h'}{m_t^2}. \tag{A.7}$$

We next compute the change in (A.7) along a ray where $x_t = \gamma m_t$ or $c_t + h(c_t/m_t) = \gamma m_t$:

$$\frac{\mathrm{d}c_t}{\mathrm{d}m_t} = \left(\frac{c_t}{m_t}\right)\left(1 + \frac{h}{c_t} + \frac{h'}{m_t}\right)\Big/\left(1 + \frac{h'}{m_t}\right). \tag{A.8}$$

Then the total change in (A.7) using (A.8) is

$$\frac{\mathrm{d}}{\mathrm{d}m_t}\left(\frac{W_m}{W_x}\right)=\left(\xi'-\frac{ch'}{h^2}\right)\left(\frac{ch}{m^3}\right)\left(\frac{c}{h}+\xi\right)^{-1}, \tag{A.9}$$

where $\xi(c_t/m_t)=(c_t/m_t)h'/h$. From (8) this expression is negative, which means that the Engel curves (joining points of constant W_m/W_x) have a smaller slope relative to the x_t axis than a ray through the origin.

To investigate the concavity of V, define $v_t\equiv U(c_t)$ and $\psi(v_t,m_t)\equiv\mu(v_t)+\phi[\mu(v_t),m_t]$, where μ is the inverse function of U. It can be shown that V is concave if and only if ψ is convex. Note that $U'>0$, $U''\leq 0$ imply $\mu'>0$, $\mu''\geq 0$. For $\phi(c_t,m_t)=h(c_t/m_t)$ satisfying (8) we can compute $\psi_{vv}\geq 0$, $\psi_{mm}\geq 0$, and

$$\begin{aligned}(\psi_{vv}\psi_{mm}-\psi_{vm}^2)=\Big\{&(h'/m_t^2)^2[2\mu\mu''-(\mu')^2]\\&+\mu^2\mu''h''(h'/m+1)+2m_t\mu\mu''h'\Big\}.\end{aligned} \tag{A.10}$$

A sufficient condition to have (A.10) non-negative is that $2\mu\mu''\geq(\mu')^2$, which is equivalent to $-c_tU''/U'\geq\frac{1}{2}$ as seen by differentiating $\mu[U(c_t)]=c_t$. Thus, V is concave for $R_u\geq\frac{1}{2}$. If U is linear so $\mu''=0$, then (A.10) is negative, so in that case V and W are proportional and are not concave, proving part (b). Q.E.D.

A.3. Proof of Proposition 3

From Proposition 1 we obtain ϕ which is equivalent to W. It can be verified from (16′) that ϕ will be homogeneous of degree one. For $\theta<\bar{\theta}$ we have $W_m>0$ which implies $\phi>0$, $\phi_m>0$ and $-\phi_c/\phi_m>m_t/c_t$, where the latter condition follows from $\phi=\phi_c c_t+\phi_m m_t$.

We wish to solve for a density function f such that (11) holds. Differentiate (11) to obtain

$$-\phi_c/\phi_m=\left[\int_{m_t/c_t}^{\bar{\theta}}\theta f(\theta)\,\mathrm{d}\theta\right]\Big/\left[\int_{m_t/c_t}^{\bar{\theta}}f(\theta)\,\mathrm{d}\theta\right]. \tag{A.11}$$

Denote the left side of (A.11) as $g(\theta)$ with $\theta\equiv m_t/c_t$. Then, for $\theta<\bar{\theta}$, we have $g>0$ and $g(\theta)>\theta$, as discussed above. From (v) in the proof of Proposition 1 we see that $W_{mm}<0$ implies that ϕ is strictly quasi-convex, so $g'(\theta)>0$. Differentiating (A.11) and rewriting we have

$$g'(\theta)\int_\theta^{\bar{\theta}}f(z)\,\mathrm{d}z-g(\theta)f(\theta)=-\theta f(\theta).$$

Dividing this equation by $g'(\theta)$ and defining $h(\theta)=[g(\theta)-\theta]/g'(\theta)>0$ for $\theta<\bar{\theta}$, we obtain

$$h(\theta)f(\theta)=\int_{\theta}^{\bar{\theta}}f(z)\,\mathrm{d}z.$$

Differentiating with respect to θ and rewriting, we obtain

$$f'(\theta)/f(\theta)=-[1+h'(\theta)]/h(\theta).$$

This differential equation is solved as

$$f(\theta)=A\exp\left\{\int_{\theta}^{\bar{\theta}}[1+h'(z)]\,\mathrm{d}z/h(z)\right\}, \tag{A.12}$$

where A is a constant. The solution (A.12) applies for $\theta\geq 0$. Note that, from (11),

$$W_m/W_x=-\phi_m=\int_{m_t/c_t}^{\bar{\theta}}f(\theta)\,\mathrm{d}\theta,$$

and since W_m/W_x is assumed to be bounded above, then so is this integral. Then A can be chosen in (A.12) along with $f(\theta)$ for $-1<\theta<0$, so that

$$\mathrm{E}\,\theta=0 \quad \text{and} \quad \int_{\underline{\theta}}^{\bar{\theta}}f(\theta)\,\mathrm{d}\theta=1. \quad \text{Q.E.D.}$$

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